

PREDICTIVE CHURN ANALYSIS USING MACHINE LEARNING

1. Introduction

Customer churn refers to the situation in which a customer stops using the products or services provided by a company. Customer retention is important for businesses because losing existing customers can affect revenue and long-term growth. Therefore, predicting customers who are likely to leave can help organizations take preventive actions.

This project, titled “**Predictive Churn Analysis**”, uses machine learning to predict customer churn based on customer demographic, subscription, billing, contract, support, and payment information.

A **Logistic Regression** classification model is developed to classify customers into two categories:

- **No Churn (0)** – the customer is expected to remain with the company.
- **Churn (1)** – the customer is expected to leave the company.

The analysis is implemented using Python in Google Colab. The dataset contains 15 customer records and 11 columns. Data preprocessing, feature transformation, model training, prediction, and evaluation are performed as part of the project.

2. Objectives

The main objectives of this project are:

1. To load and examine a customer churn dataset.
2. To check the dataset for missing values and duplicate records.
3. To analyze the distribution of the churn target variable.
4. To separate input features from the target variable.
5. To identify numerical and categorical features.
6. To preprocess the data using One-Hot Encoding and MinMax Scaling.
7. To divide the dataset into training and testing sets.
8. To develop a Logistic Regression model for churn prediction.
9. To calculate churn probabilities for test customers.
10. To evaluate the model using Precision, Recall, F1-Score, ROC-AUC, and a Confusion Matrix.

3. Dataset Description

The project uses a customer churn dataset named `customer_churn_sample.csv`.

The dataset contains:

- **15 records**
- **11 columns**

The columns included in the dataset are:

Column	Description
CustomerID	Unique identification number of the customer
Gender	Gender of the customer
Age	Age of the customer
TenureMonths	Number of months the customer has been associated with the service
SubscriptionType	Type of subscription
MonthlyCharges	Monthly amount charged to the customer
TotalCharges	Total amount charged to the customer
ContractType	Type of customer contract
SupportTickets	Number of support tickets raised
PaymentMethod	Method used for payment
Churn	Indicates whether the customer churned

The **Churn** column is the target variable for the machine learning model.

4. Data Loading and Initial Analysis

The dataset was loaded using the Pandas library.

The initial analysis showed that the dataset contains **15 rows and 11 columns**. The notebook also checked for missing values and duplicate records.

The results were:

- Missing values: **0 in all columns**
- Duplicate rows: **0**
- Churn = No: **8 customers**
- Churn = Yes: **7 customers**

Therefore, the dataset did not contain missing or duplicate records that required additional cleaning.

The distribution of the target variable was:

Churn Status	Number of Customers
No	8
Yes	7
Total	15

This shows that the dataset has a relatively balanced distribution between the two churn classes.

5. Data Preprocessing

5.1 Target Variable Conversion

The original target variable contained the categorical values “**Yes**” and “**No**”.

These values were converted into numerical form:

- No → 0
- Yes → 1

This conversion allows the classification algorithm to work with the target variable numerically.

5.2 Feature Selection

The **CustomerID** column was removed because it is an identifier rather than a meaningful predictive feature.

The **Churn** column was separated as the target variable.

The input features used by the model were:

- Gender
- Age
- TenureMonths
- SubscriptionType
- MonthlyCharges
- TotalCharges
- ContractType
- SupportTickets
- PaymentMethod

The target variable was:

- Churn

5.3 Numerical Features

The following variables were treated as numerical features:

- Age
- TenureMonths
- MonthlyCharges
- TotalCharges
- SupportTickets

These features represent measurable numerical information about each customer.

5.4 Categorical Features

The following variables were treated as categorical features:

- Gender
- SubscriptionType
- ContractType
- PaymentMethod

Since machine learning models generally require numerical inputs, these categorical variables were transformed into numerical representations.

5.5 One-Hot Encoding

One-Hot Encoding was applied to the categorical features.

The preprocessing pipeline used:

- OneHotEncoder
- handle_unknown="ignore"
- drop="first"

One-Hot Encoding converts categorical values into binary indicator variables so that they can be processed by the Logistic Regression algorithm.

5.6 MinMax Scaling

MinMax Scaling was applied to the numerical features.

The purpose of scaling is to transform numerical values to a common range, reducing the effect of differences in the original scales of variables such as Age, MonthlyCharges, and TotalCharges.

The notebook combines the numerical scaling and categorical encoding using a **ColumnTransformer**.

6. Train-Test Split

The dataset was divided into training and testing data using an **80/20 train-test split**.

The split was performed using:

- Test size: 20%
- Random state: 42
- Stratification: Applied

Since the dataset contains 15 records, the resulting split was:

Dataset	Number of Records
Training set	12
Testing set	3

Dataset	Number of Records
Total	15

The training set contained 6 No-Churn and 6 Churn customers, while the testing set contained 2 No-Churn and 1 Churn customer.

7. Machine Learning Model

7.1 Logistic Regression

Logistic Regression was selected as the classification algorithm.

Logistic Regression is commonly used for binary classification problems. In this project, the model predicts one of two possible outcomes:

- 0 → No Churn
- 1 → Churn

The model was implemented as a pipeline containing:

1. Data preprocessing
2. Logistic Regression classifier

The model was configured with:

- `max_iter = 1000`
- `random_state = 42`

The model was trained using the training dataset.

8. Churn Prediction

After training, the model was used to predict the classes of customers in the test dataset.

The predicted classes were:

[0, 1, 0]

This means that the model predicted:

- Customer/Test Record 1 → No Churn
- Customer/Test Record 2 → Churn
- Customer/Test Record 3 → No Churn

The model also calculated the probability of churn for each test record:

Test Record	Churn Probability
1	0.1007

Test Record	Churn Probability
2	0.7893
3	0.3403

The second test record had the highest predicted churn probability of approximately **78.93%**.

9. Model Evaluation

The performance of the Logistic Regression model was evaluated using several classification metrics.

9.1 Precision

Precision measures how many of the customers predicted as churners were actually churners.

The obtained Precision was:

1.00

or

100%

9.2 Recall

Recall measures how many of the actual churn customers were correctly identified by the model.

The obtained Recall was:

1.00

or

100%

9.3 F1-Score

The F1-Score combines Precision and Recall into a single measure.

The obtained F1-Score was:

1.00

or

100%

9.4 ROC-AUC

The ROC-AUC score evaluates the model's ability to distinguish between the two classes.

The obtained ROC-AUC score was:

1.00

This indicates perfect separation of the classes on the available test set.

9.5 Performance Summary

Metric	Score
Precision	1.00
Recall	1.00
F1-Score	1.00
ROC-AUC	1.00
Accuracy	1.00

The notebook's classification report also shows 1.00 for both No Churn and Churn classes, with a test support of 2 and 1 respectively.

10. Confusion Matrix

A confusion matrix was generated to visualize the classification results.

The resulting confusion matrix was:

	Predicted No Churn	Predicted Churn
Actual No Churn	2	0
Actual Churn	0	1

Therefore:

- True Negatives = 2
- False Positives = 0
- False Negatives = 0
- True Positives = 1

The model correctly classified all three test records.

11. ROC Curve

A Receiver Operating Characteristic (ROC) curve was generated to evaluate the classification performance of the Logistic Regression model.

The curve compares:

- **True Positive Rate**
- **False Positive Rate**

The calculated **AUC was 1.00**. The notebook also plots a diagonal reference line representing a random classifier.

A higher AUC generally represents better discrimination between the two classes. In this experiment, the obtained value of 1.00 indicates that the model perfectly separated the test examples.

12. Results and Discussion

The Logistic Regression model produced excellent evaluation results on the available test dataset.

The model achieved:

- 100% Precision
- 100% Recall
- 100% F1-Score
- 100% ROC-AUC
- 100% accuracy on the three test records

The confusion matrix also shows that all three test records were correctly classified.

The model additionally provides churn probabilities rather than only class labels. For example, one of the test records received a churn probability of approximately **0.789**, indicating a relatively high predicted likelihood of churn.

However, the dataset contains only **15 customer records**, with only **3 records in the test set**. Therefore, the perfect scores should not be interpreted as proof that the model would achieve 100% performance on a larger real-world dataset. A larger dataset would be required for a more reliable assessment.

13. Applications of Customer Churn Prediction

A churn prediction system can be useful for organizations in several ways:

1. **Customer Retention**

Businesses can identify customers who may leave and take appropriate retention measures.

2. **Targeted Offers**

Customers with higher churn probabilities can potentially receive suitable offers or incentives.

3. **Customer Support Improvement**

Customer information such as support tickets can be analyzed to understand customer behavior.

4. **Business Decision-Making**

Churn predictions can help management make data-driven customer retention decisions.

5. Revenue Protection

Early identification of potential churn can help businesses reduce customer losses.

14. Technologies and Libraries Used

The project was implemented using the following technologies:

Technology/Library	Purpose
Python	Programming language
Google Colab	Development environment
Pandas	Dataset loading and analysis
Scikit-learn	Machine learning and evaluation
Matplotlib	Visualization
Logistic Regression	Churn classification
ColumnTransformer	Feature preprocessing
OneHotEncoder	Categorical feature encoding
MinMaxScaler	Numerical feature scaling

The notebook uses a preprocessing pipeline to combine data transformation and model training.

15. Limitations

Although the project produced excellent results, there are several limitations:

1. Very small dataset

The dataset contains only 15 records.

2. Small test set

Only 3 records were used for testing, so the evaluation metrics are based on a very limited number of examples.

3. Limited generalization

The perfect scores may not represent performance on a larger and more diverse customer dataset.

4. Limited analysis of feature importance

The current notebook focuses primarily on prediction and model evaluation and does not include a detailed analysis of which individual features contribute most to churn.

5. Synthetic/sample nature of the dataset

The file is named `customer_churn_sample.csv`, so the results should be treated as a demonstration of the machine learning workflow rather than a production-ready churn prediction system.

16. Future Scope

The project can be improved in the future by:

- Using a much larger customer dataset.
- Comparing Logistic Regression with other classification algorithms such as Decision Tree, Random Forest, and Support Vector Machine.
- Applying cross-validation for more reliable evaluation.
- Performing hyperparameter tuning.
- Analyzing feature importance and customer churn patterns.
- Creating an interactive dashboard.
- Developing a system that automatically identifies high-risk customers.
- Using real-world customer data to validate the model.

17. Conclusion

The **Predictive Churn Analysis** project demonstrates the use of machine learning for binary customer churn prediction. The dataset was successfully loaded and checked for missing values and duplicate records. The target variable was converted into numerical form, while numerical and categorical features were processed using MinMax Scaling and One-Hot Encoding.

An 80/20 stratified train-test split was used, resulting in 12 training records and 3 testing records. A Logistic Regression model was then trained and used to predict customer churn.

The model achieved **1.00 Precision, 1.00 Recall, 1.00 F1-Score, and 1.00 ROC-AUC**, while the confusion matrix showed that all three test records were classified correctly.

Overall, the project successfully demonstrates the complete basic workflow of a machine learning classification problem, from dataset loading and preprocessing to model development, prediction, and evaluation. However, because the dataset is very small, further testing with a larger dataset is necessary before making conclusions about real-world predictive performance.

18. References

1. Python documentation and standard Python libraries.
2. Pandas library for data manipulation and analysis.
3. Scikit-learn library for machine learning and model evaluation.
4. Matplotlib library for data visualization.
5. Project dataset: `customer_churn_sample.csv`.
6. Google Colab notebook developed for Predictive Churn Analysis.